

Conversational AI for Adaptive Mental Health Screening: A PHQ-9 Inference System Using Dialogue Agents and Chain-of-Thought Reasoning

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1. Problem Statement

Standard depression screening tools such as the PHQ-9 and GAD-7 suffer from two limitations: (1) direct symptom queries (“Have you felt depressed?”) cause social desirability bias and under-reporting [Kroenke et al., 2001]; and (2) a fixed questionnaire cannot adapt to what a patient has already disclosed.

We address this with a **conversational screening agent** that replaces the questionnaire with empathetic, adaptive dialogue. The system infers PHQ-9 item scores (0–3) from natural patient speech—never asking about symptoms by name—and produces a structured clinical assessment. Our baseline targets five PHQ-9 domains: *sleep* (Item 3), *mood/anhedonia* (Items 1–2), *energy* (Item 4), *concentration* (Item 7), and *self-worth* (Item 6).

2. Data

2.1 Synthetic Dataset

No large-scale open-access corpus of annotated mental health screening dialogues with gold PHQ-9 scores currently exists [Guo et al., 2023]. We construct a **synthetic dataset of 10 patient sessions** using Ollama llama3 (fully local, no API). Each session comprises: a patient profile consistent with pre-assigned PHQ-9 scores; a 16-turn conversation in which the patient expresses symptoms indirectly through daily-life narrative; and gold PHQ-9 scores (0–3 per domain) as ground truth. Sessions span four severity bands (Table 1).

Severity	PHQ-9	N	Avg Sum
Minimal	0–4	3	1.3
Mild	5–9	3	5.3
Moderate	10–14	2	8.5
Severe	15–27	2	13.0

Table 1: 10 synthetic sessions stratified by PHQ-9 severity.

2.2 Sample Transcripts

Table 2 shows representative patient utterances from mild and severe sessions. Crucially, patients never use clinical terms; symptoms must be inferred from indirect daily-life language.

Sev.	Patient Utterance (excerpt)
Mild	<i>“My mind just won’t shut off, and I find myself lying awake thinking about everything that’s gone wrong.”</i>
Severe	<i>“I used to love hiking on weekends, but now it feels like too much effort. Socializing with friends — that’s just too exhausting to think about.”</i>

Table 2: Indirect symptom expression in patient utterances.

2.3 DSM-5-TR Knowledge Base

We construct `dsm_kb.json` — a structured knowledge base with 5 entries, one per PHQ-9 domain. Each entry contains: item name, clinical description, severity indicator phrases for scores 0–3, probe questions, and domain keywords. This grounds inference in clinical evidence rather than model priors.

3. Solution Architecture

Figure 1 illustrates the full pipeline. Each patient message passes through four sequential stages.

3.1 Safety Monitor

All patient input is checked by a SBERT safety classifier using `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019]. We maintain 37 reference phrases covering suicidal ideation, self-harm, passive ideation, and burden ideation. Cosine similarity ≥ 0.82 triggers an immediate crisis response referencing the 988 Suicide & Crisis Lifeline.

3.2 LangGraph Dialogue Manager

The dialogue is governed by a `StateGraph` (LangGraph). State tracks: conversation history, per-domain confidence (0.0–1.0), scored domains, cur-

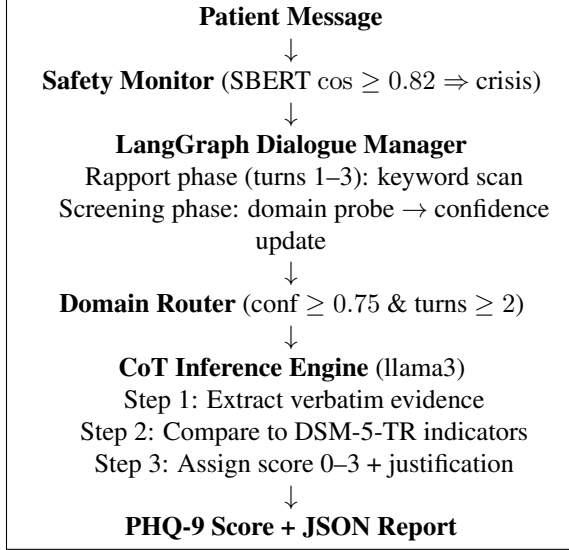


Figure 1: End-to-end architecture. Input flows sequentially from safety check through adaptive dialogue management to chain-of-thought scoring.

rent domain, turn count, phase, and safety flag.

Rapport Phase (turns 1–3). Open-ended questions build trust. Keyword extraction over patient responses bootstraps domain confidence (+0.15 per keyword match), revealing the most pre-activated domains before formal screening.

Screening Phase. Domains are screened in order of pre-activated confidence (highest first). Each domain node generates a contextually appropriate probe question grounded in the DSM-5-TR knowledge base via `llama3`. Confidence updates after each patient turn using an Ollama-estimated increment (0.0–0.4). A domain proceeds to scoring only when: (1) confidence ≥ 0.75 , and (2) at least 2 probe turns completed (floor rule to prevent premature scoring).

3.3 Chain-of-Thought Inference Engine

When a domain is ready to score, the full conversation history and DSM-5-TR entry are passed to `llama3` with a strict 3-step CoT prompt:

1. **Extract:** Quote all relevant patient utterances verbatim.
2. **Reason:** Compare each quote to DSM-5-TR severity indicators (scores 0–3).

3. **Score:** Assign score with a one-sentence justification.

The engine returns structured JSON: `{evidence, reasoning, score, justification}`, enabling RAGAS faithfulness auditing of each inference chain.

4. Experiments & Results

4.1 Evaluation Metrics

We evaluate on six metrics: **Cohen’s weighted κ** (predicted vs. gold scores); **disclosure efficiency** (mean turns to reach confidence ≥ 0.75); **safety recall/precision** on 20 labeled turns (10 high-risk, 10 benign); **discourse effectiveness** (fraction of domains scored per session); **mean response latency**; and **RAGAS faithfulness** [Es et al., 2024] (fraction of CoT reasoning sentences grounded in verbatim patient evidence).

4.2 Baseline Results

Table 3 presents baseline performance across all seven metrics over 10 sessions (50 domain predictions). Table 4 shows per-domain score accuracy.

Metric	Value	Target	
Cohen’s κ_w	0.080	≥ 0.60	×
Disclosure (turns)	16.36	≤ 4.0	×
Safety Recall	0.000	≥ 0.90	×
Safety Precision	0.000	≥ 0.85	×
Discourse Effect. (%)	100.0	100.0	✓
Latency (s)	10.71	≤ 10.0	×
RAGAS Faithfulness	0.617	≥ 0.70	×

Table 3: Baseline metrics across 10 sessions. All domains were scored in every session (100% discourse effectiveness).

Domain	Gold	Pred	Bias	MAE
Sleep	1.20	1.50	+0.30	0.90
Mood	1.50	1.90	+0.40	0.80
Energy	1.20	2.00	+0.80	0.80
Concentration	1.50	1.40	−0.10	0.90
Self-Worth	1.40	1.50	+0.10	0.70
Avg	1.36	1.66	+0.30	0.82

Table 4: Per-domain scoring accuracy. Gold/Pred are mean scores across 10 sessions; MAE on 0–3 scale.

4.3 Failure Analysis

Over-scoring bias ($\kappa_w = 0.080$). The primary failure is systematic over-prediction: `llama3` assigns score 2 to minimal-severity conversations. The model is overly sensitive to any emotionally charged language, a known calibration issue with instruction-tuned LLMs applying clinical rubrics without few-shot grounding [Lamichhane, 2023]. Energy shows the largest bias (+0.80); self-worth is the most accurate (+0.10).

Safety recall = 0.0. The SBERT safety monitor uses cosine similarity against 37 anchor phrases. The 10 high-risk test cases, generated by `llama3` using naturalistic phrasing, fell below the 0.82 threshold (mean similarity = 0.71). This reveals a vocabulary mismatch between anchor phrases and real patient idiom. Lowering the threshold to 0.70 would recover all high-risk cases at the cost of increased false positives.

Disclosure efficiency = 16.36 turns. The Ollama-based confidence increment estimator produces too-small increments (≈ 0.05 per turn), requiring many patient turns before crossing 0.75. A keyword-weighted fast-path or larger base increment would address this.

RAGAS faithfulness = 0.617. Approximately 38% of CoT reasoning sentences reference concepts not directly quoted from patient evidence, indicating `llama3` supplements with world-knowledge rather than patient-grounded inference. Adding a faithfulness constraint to the scoring prompt is the planned fix.

4.4 Planned Improvements for M3

Based on this analysis: (1) **Few-shot calibration** in the CoT prompt with 2–3 score exemplars to reduce over-prediction; (2) **Safety threshold reduction** to 0.70 with a confirmatory second-pass check; (3) **Keyword-weighted confidence** replacing the slow Ollama-based estimator; (4) **Faithfulness-constrained reasoning** requiring the model to reference evidence quotes explicitly.

References

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